# Computer Science GCE A Level 2024 - Paper 1

Duration: 1h30 | Cameroon GCE Board

GENERAL CERTIFICATE OF EDUCATION BOARD, General Certificate of Education Examination. 0795 COMPUTER SCIENCE 1, ADVANCED LEVEL, JUNE 2024. Multiple choice question paper. Duration: one and a half hours. Mobile phones and calculators are NOT allowed in the examination room. Answer ALL the 50 questions in this examination. All questions carry equal marks. Each question has FOUR suggested answers: A, B, C and D. Decide which answer is correct, find the number of the question on the answer sheet and draw a horizontal line across the letter to join the square brackets for the answer you have chosen. Mark only one answer for each question (if more than one answer is marked, the question scores zero). Avoid spending too much time on any one question. Do all rough work in the booklet.

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**1.** It uses the same instruction sequences to process in the same way large amounts of data streamed to it. At best it must be:

a) Multiple-instruction, single-data.
b) Multiple-instruction, multiple-data.
c) Single-instruction, single-data.
d) Single-instruction, multiple-data.

**2.** Suh has data to process, at the same time, on powerful machines in various parts of the world. It is best he does this on a(n):

a) Online Computer.
b) Distributed Computer.
c) Mainframe Computer.
d) Parallel Computer.

**3.** We seek it for long-term storage, temporary storage for executing instructions and data, and for sheer bulk. It is a:

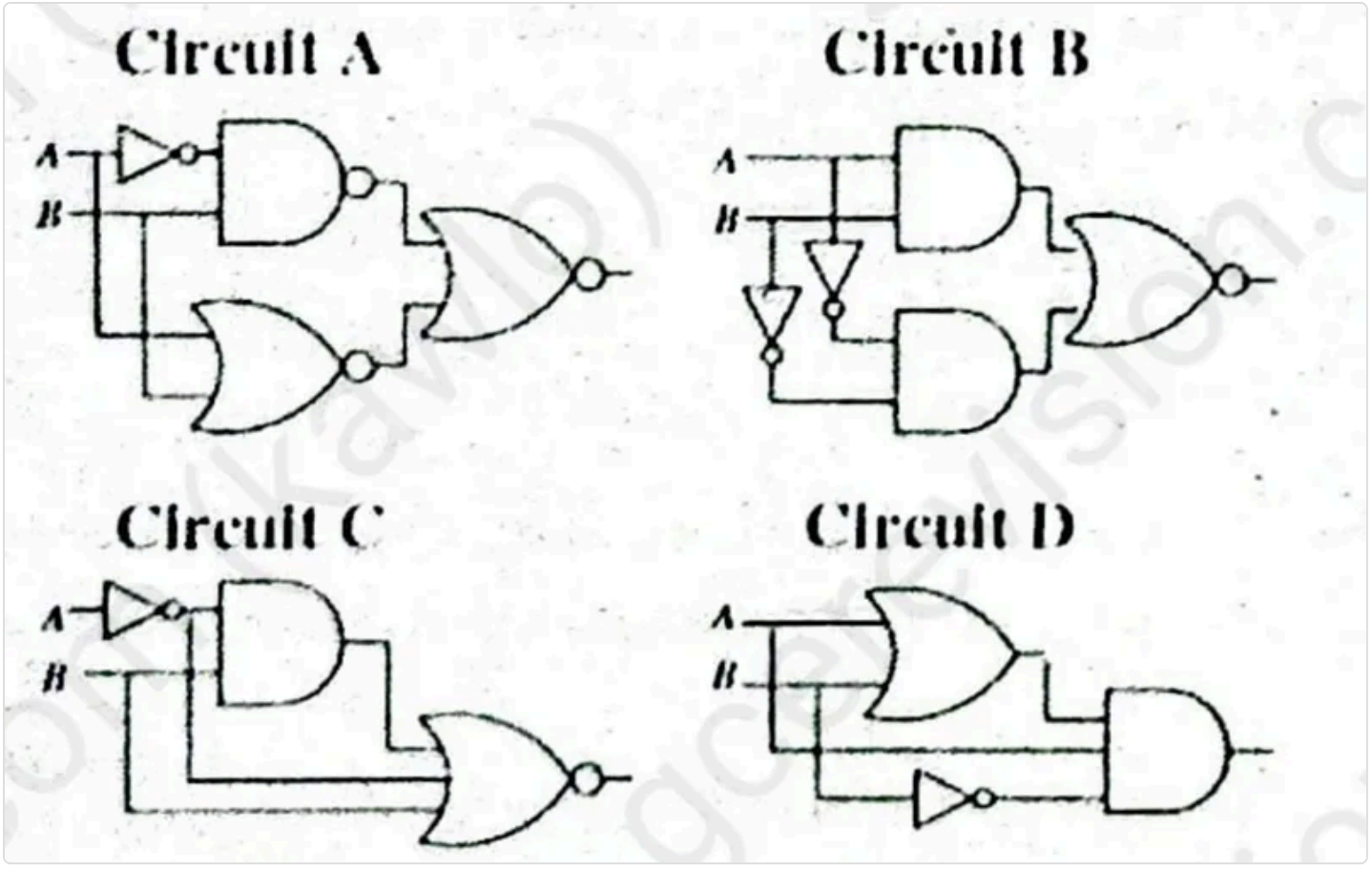
a) Main memory.
b) Hard drive.
c) Cache memory.
d) USB drives.

**4.** What is the result of the following in ones complement notation?

+10 110 011 - 10 001 010

a) 00 101 001.
b) 00 101 000.
c) 10 101 001.
d) 10 011 000.

**5.** Which of the following circuits corresponds to a reexpression of the Boolean formula $\overline{(A + \bar{B}) + \overline{A + B}}$ ?




a) Circuit A.
b) Circuit B.
c) Circuit C.
d) Circuit D.

**6.** Which of the following corresponds to the truth table adjacent?

| A | B | R |
|---|---|---|
| F | F | T |
| F | T | T |
| T | F | T |
| T | T | F |

a) $\bar{A} \cdot \bar{B} + \overline{A \cdot B}$.
b) $A \cdot \overline{B + A} \cdot B$.
c) $\bar{A} \cdot B + A \cdot B$.
d) $\overline{A \cdot B + \overline{A \cdot B}}$.

**7.** What guides a machine on how to execute any instruction given it?

*Options as printed:* A) The values to use in operations. B) The machine addressing mode. C) The machine instruction set. D) The instruction format.

**8.** A machine's CPU takes arguments for its binary operations from the accumulator. It surely is a:

a) Three-address machine.
b) Two-address machine.
c) One-address machine.
d) Zero-address machine.

**9.** In indexed mode addressing of machines:

a) The operand of the instruction is not used immediately.
b) The operand is an offset to the location where the sought value is found.
c) The operand field holds an address for its immediate use.
d) The operand address is given as an offset from the instruction.

**10.** It is used for work, for teaching and learning, without preliminary training, for instant feedback and suitable displays. It must be a(n):

a) Video Conferencing System.
b) Email system.
c) Information retrieval system.
d) Office automation system.

**11.** It is good at repetitive tasks, a good replacement for humans in hostile environments. It must be:

a) Computer-aided Manufacturing.
b) An intelligent machine.
c) A Simulation Game.
d) A robot.

**12.** It is good for incidents reporting, tracking evolution over time, across distances, and must manage large amounts of data.

a) Database Management Systems.
b) Information & Communication Systems.
c) Geographic Information Systems.
d) Management Information Systems.

**13.** In handling data, which of the following is NOT a breach of privacy?

a) Let Zia verify data Joe must have input.
b) Ndi forgot his medical records in a park.
c) Obey a court order on Fuh's debts.
d) Idi makes public Pat's medical record.

**14.** Which of the following is a computer threat that affects content owned by another?

*Options as printed:* A) Flood a website with service requests. B) Illegal change to someone's password. C) Spoofing where data is sent in a network. D) Corruption of a file system in a network.

**15.** Which of these terms from the early days of operating systems is most correctly defined:

*Options as printed:* A) Batch systems grouped tasks and executed them by a single user. B) Multi-tasking systems carry out multiple tasks. C) Multi-user systems shared machine resources among users at the same time. D) Real-time systems were interactive in real, world-clock time.

**16.** Temporal databases record:

a) Objects and their relationships as they evolve over time.
b) Objects and their relationships at fixed points in time.
c) Record changing values and relationships at fixed positions in time.
d) Record the values of objects as their relationships change in time.

**17.** It enables the independent reuse of program-level functionalities in other programs as they are readied for execution. They are:

a) Utility programs.
b) Library programs.
c) Application programs.
d) Systems programs.

**18.** Which of the following compiler phases best work together so that strings of symbols are given the meaning expected of the program when executed.

a) Lexical analysis and syntax analysis.
b) Syntax analysis and code generation.
c) Code generation and code optimisation.
d) Code optimisation and code execution.

**19.** Mbu, a systems administrator just wants to navigate Google Earth to locate some information for friends. On the other hand, Ngwe is new to systems administration and wants to learn. It is best for Mbu and Ngwe respectively to use:

a) MS Windows GUI; MS Windows GUI.
b) MS Windows GUI; Linux CLI.
c) Linux CLI; Linux CLI.
d) Linux CLI; MS Windows GUI.

**20.** Which of the following makes it easiest to swap locally executing code into memory?

*Options as printed:* A) Variable memory partitioning. B) Paged memory allocation. C) Segmented memory allocation. D) Segmented demand-paged allocation.

**21.** In what circumstance(s) would it be best to use a Shortest Job First policy?

*Options as printed:* A) In order to avoid CPU preemption. B) In order to reduce CPU waiting time. C) When there are many short jobs and it is not clear which ones to consider first. D) When it is important to improve on the overall turnaround time of processes.

**22.** In which of the following scenarios would normalisation NOT be of help?

a) Record Che's library registration without him having to borrow at least one book.
b) Remove Njoh as a library user, without deleting his library usage history.
c) Awaiting Pat to update her CV since she became Chief librarian.
d) Update the details of Egbe found in many activities in her university.

**23.** In relational databases which of the following is true?

a) Candidate keys can be used as primary keys.
b) A key cannot include all attributes.
c) Foreign keys cannot be candidate keys.
d) Candidate keys of a relation can be null.

**24.** It allows one to access, update and otherwise manipulate the data found in a DBMS:

a) Data Definition Language (DDL).
b) Data Manipulation Language (DML).
c) Structured Query Language (SQL).
d) Fourth Generation Language (4GL).

**25.** Which of the following is most conveniently used to specify the requirements of a program to develop?

*Options as printed:* A) Dataflow diagrams. B) E-R diagrams. C) UML charts. D) Flow charts.

**26.** It is used to make sure that management and others have needed data in tailored and well-presented forms. All in the organisation can acquire its skills.

*Options as printed:* A) Document preparation. B) Laptop computing. C) Job retraining. D) Office automation.

**27.** It lets data from channels go over long distances, before possibly being resent to other devices:

a) Multiplexer.
b) Repeater.
c) Modem.
d) Codec.

**28.** It lets one forward data to other networks in an internet, no matter how distant they are:

a) Router.
b) Gateway.
c) Bridge.
d) Repeater.

**29.** Bisuh receives data from various parts of the world, in varying quantities and rates and, most critically, only when she is ready to handle the data. Data transmission must be:

*Options as printed:* A) signalled. B) buffered. C) synchronised. D) coordinated.

**30.** In relation to database security, which of the following does NOT detect or prevent threats?

a) Data encryption.
b) Data backup.
c) Integrity constraints.
d) Standards enforcement.

**31.** She physically realises the design, implementation, security and controls of the database system as well as the maintenance and performance of the operational system. She is certainly a:

a) Application Developer.
b) Database Designer.
c) Database Administrator.
d) Data Administrator.

**32.** Assuming identifier names are of obvious meanings, the SQL query: `SELECT author FROM booklist WHERE published >= 5 AND livesIn = 'Kom'`.

a) List authors in Kom who published 5 or more books.
b) Selects from the file 'author' all entries having 'Kom' and at least 5 published.
c) List book authors of at least 5 books that live in Kom.
d) Identify authors that have at least 5 books published or live in Kom.

**33.** Which of these statements is false? A relation is in second normal form (2NF) if:

a) No attribute can depend on non-key attributes.
b) All attributes are fully dependent on the primary key.
c) Candidate key attributes can transitively depend on the primary key.
d) A primary key attribute may fully depend on the primary key.

**34.** Computing arithmetic expressions (from left to right) has to check operator precedence such as BEDMAS. To avoid this and execute them in the correct precedence order requires transforming the expressions using:

a) Pre order traversal.
b) Post order traversal.
c) In order traversal.
d) Mixed order traversal.

**35.** Which of these data types is least suited to repeatedly process data in the same way?

*Options as printed:* A) File. B) List. C) Array. D) String.

**36.** In a small restaurant service, current clients, the day's menu and drinks options are each given a 4-bit code. If half the client ID's, menu and drinks options are currently available, how many client-food-drink triples can still be encoded?

a) 512.
b) 256.
c) 64.
d) 128.

**37.** A software application provides services and resources identified to mostly self-contained subtasks, with explicit interfacing for use of services and resources. The best design strategy to use is:

*Options as printed:* A) Services design. B) Modular design. C) Top-down design. D) Bottom-up design.

**38.** An algorithm initially sets variables $r$ and $s$ 10 and 0. If $r \geq 2$ it then subtracts 3 from $r$ and adds 2 to $s$, and repeatedly does so until $r

*Options as printed:* A) 0. B) 6. C) 14. D) 16.

**39.** What are the values of x and y just after h is last called, if its first call is h(10,17)?

*Options as printed:* A) 3, 7. B) 10, 7. C) 3, 3. D) 3, 4.

**40.** Two variables in a code fragment can have the same name and be used without confusion because they have different:

a) Locations.
b) Declarations.
c) Scope.
d) Syntax.

**41.** Sina wants to make sure a system software component he updated does not impede the functioning of other existing applications. It is best Sina consults the system's:

*Options as printed:* A) Design Specification. B) Test Plan. C) Technical Manual. D) Installation Manual.

**42.** We examine executed code to establish its execution patterns in order to make suitable corrections. It is:

*Options as printed:* A) Debugging. B) Testing. C) Reviewing. D) Profiling.

**43.** In complexity theory, a hard problem is/has:

a) A polynomial time complexity.
b) An exponential time complexity.
c) Too hard for current computers to solve.
d) Known to have no solution whatsoever.

**44.** In computability, the term semi-decidable means that in general a problem:

a) Sometimes has a solution.
b) Has a solution half the time.
c) Does have a solution if there is one.
d) Does not yet have a solution.

**45.** It emphasises the production of a working system that achieves the goals of the software. The activity must be:

a) Software design.
b) Software testing.
c) Software implementation.
d) Software installation.

**46.** A high-level feature most convenient to use if a value is occasionally changed in implementations of a program is:

a) Constant declaration.
b) Variable declaration.
c) Parameter declaration.
d) Function argument.

**47.** Different components of a task could be executed independently at the same time, though on much fewer processors. For this, we designed a(n):

a) Concurrent algorithm.
b) Parallel algorithm.
c) Distributed algorithm.
d) Cloud Algorithm.

**48.** It has entities that interact via clear rules to communicate with each other, and to apply their influences on each other. The best programming language paradigm to use is:

a) Object-oriented paradigm.
b) Functional paradigm.
c) Declarative paradigm.
d) Imperative paradigm.

**49.** In programs, it holds values we access and is called by different names. But when we use it to communicate between self-contained program units, we must call it:

a) A variable.
b) A parameter.
c) An argument.
d) A location.

**50.** We examine software to make sure it conforms to the customer's specified requirements. It must be:

a) Black box testing.
b) White box testing.
c) Integration testing.
d) Acceptance testing.


### Want to check your answers?

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